MOTOROLA Semiconductors
BOX 20912 • PHOENIX, ARIZONA 85036

MC6800

(0 to 70°C; L or P Suffix)

MC6800C

(-40 to 85°C; L Suffix only)

MICROPROCESSING UNIT (MPU)

The MC6800 is a monolithic 8-bit microprocessor forming the central control function for Motorola's M6800 family. Compatible with TTL, the MC6800, as with all M6800 system parts, requires only one +5.0-volt power supply, and no external TTL devices for bus interface.

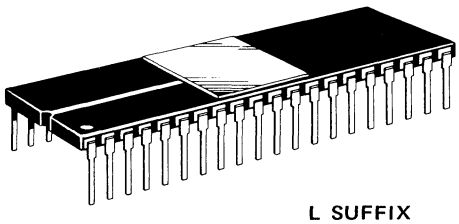
The MC6800 is capable of addressing 65K bytes of memory with its 16-bit address lines. The 8-bit data bus is bidirectional as well as 3-state, making direct memory addressing and multiprocessing applications realizable.

- Eight-Bit Parallel Processing
- Bi-Directional Data Bus
- Sixteen-Bit Address Bus — 65K Bytes of Addressing
- 72 Instructions — Variable Length
- Seven Addressing Modes — Direct, Relative, Immediate, Indexed, Extended, Implied and Accumulator
- Variable Length Stack
- Vectored Restart
- Maskable Interrupt Vector
- Separate Non-Maskable Interrupt — Internal Registers Saved In Stack
- Six Internal Registers — Two Accumulators, Index Register, Program Counter, Stack Pointer and Condition Code Register
- Direct Memory Addressing (DMA) and Multiple Processor Capability
- Clock Rates as High as 1 MHz
- Simple Bus Interface Without TTL
- Halt and Single Instruction Execution Capability

MOS

(N-CHANNEL, SILICON-GATE)

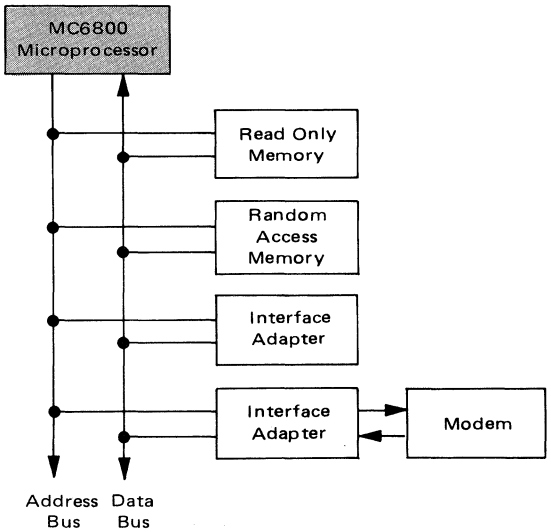
MICROPROCESSOR



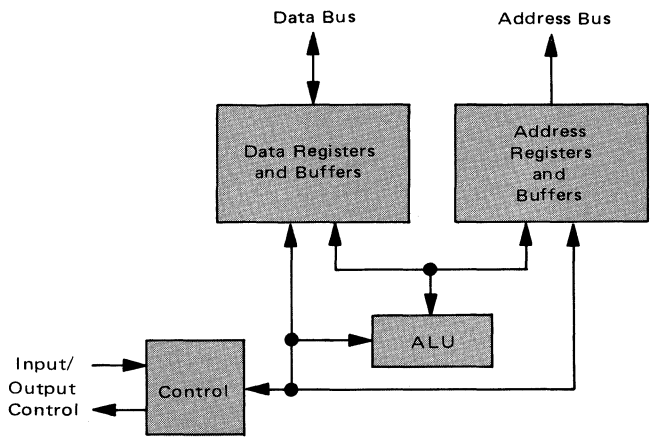
L SUFFIX
CERAMIC PACKAGE
CASE 715

NOT SHOWN: **P SUFFIX**
PLASTIC PACKAGE
CASE 711

M6800 MICROCOMPUTER FAMILY BLOCK DIAGRAM



MC6800 MICROPROCESSOR BLOCK DIAGRAM

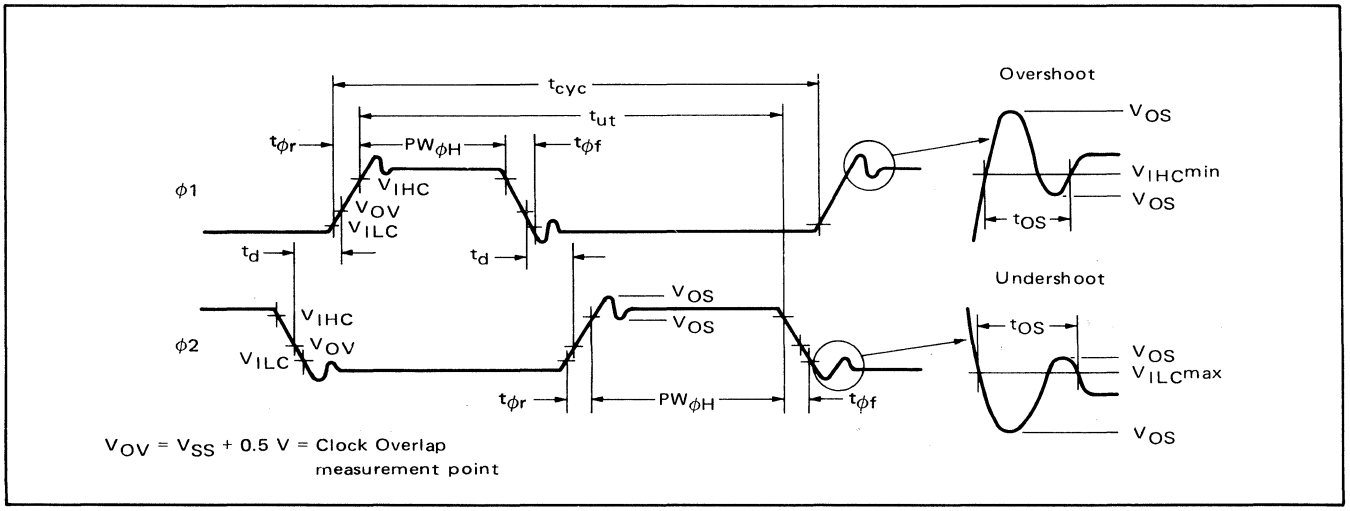


ELECTRICAL CHARACTERISTICS (V_{CC} = 5.0 V ± 5%, V_{SS} = 0, T_A = 0 to 70°C unless otherwise noted.)

Characteristic		Symbol	Min	Typ	Max	Unit
Input High Voltage	Logic	V_{IH}	V_{SS} + 2.0	–	V_{CC}	Vdc
	$\phi1,\phi2$	V_{IHC}	V_{CC} – 0.3	–	V_{CC} + 0.1	
Input Low Voltage	Logic	V_{IL}	V_{SS} – 0.3	–	V_{SS} + 0.8	Vdc
	$\phi1,\phi2$	V_{ILC}	V_{SS} – 0.1	–	V_{SS} + 0.3	
Clock Overshoot/Undershoot – Input High Level		V_{OS}	V_{CC} – 0.5	–	V_{CC} + 0.5	Vdc
– Input Low Level			V_{SS} – 0.5	–	V_{SS} + 0.5	
Input Leakage Current		I_{in}				μAdc
(V_{in} = 0 to 5.25 V, V_{CC} = max)	Logic*		–	1.0	2.5	
(V_{in} = 0 to 5.25 V, V_{CC} = 0.0 V)	$\phi1,\phi2$		–	–	100	
Three-State (Off State) Input Current	D0-D7	I_{TSI}	–	2.0	10	μAdc
(V_{in} 0.4 to 2.4 V, V_{CC} = max)	A0-A15,R/W		–	–	100	
Output High Voltage		V_{OH}				Vdc
(I_{Load} = –205 μAdc, V_{CC} = min)	D0-D7		V_{SS} + 2.4	–	–	
(I_{Load} = –145 μAdc, V_{CC} = min)	A0-A15,R/W,VMA		V_{SS} + 2.4	–	–	
(I_{Load} = –100 μAdc, V_{CC} = min)	BA		V_{SS} + 2.4	–	–	
Output Low Voltage (I_{Load} = 1.6 mAdc, V_{CC} = min)		V_{OL}	–	–	V_{SS} + 0.4	Vdc
Power Dissipation		P_D	–	0.600	1.2	W
Capacitance #	$\phi1,\phi2$	C_{in}	80	120	160	pF
(V_{in} = 0, T_A = 25°C, f = 1.0 MHz)	TSC		–	–	15	
	DBE		–	7.0	10	
	D0-D7		–	10	12.5	
	Logic Inputs		–	6.5	8.5	
	A0-A15,R/W,VMA	C_{out}	–	–	12	pF
Frequency of Operation		f	0.1	–	1.0	MHz
Clock Timing (Figure 1)						
Cycle Time		t_{cyc}	1.0	–	10	μs
Clock Pulse Width		$PW_{\phi H}$				ns
(Measured at V_{CC} – 0.3 V)	$\phi1$		430	–	4500	
	$\phi2$		450	–	4500	
Total $\phi1$ and $\phi2$ Up Time		t_{ut}	940	–	–	ns
Rise and Fall Times	$\phi1,\phi2$	$t_{\phi r}$, $t_{\phi f}$	5.0	–	50	ns
(Measured between V_{SS} + 0.3 V and V_{CC} – 0.3 V)						
Delay Time or Clock Separation		t_d	0	–	9100	ns
(Measured at V_{OV} = V_{SS} + 0.5 V)						
Overshoot Duration		t_{OS}	0	–	40	ns

*Except $\overline{IRQ}$ and $\overline{NMI}$, which require 3 kΩ pullup load resistors for wire-OR capability at optimum operation.
#Capacitances are periodically sampled rather than 100% tested.

FIGURE 1 – CLOCK TIMING WAVEFORM



MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	−0.3 to +7.0	Vdc
Input Voltage	V_{in}	−0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	−55 to +150	°C
Thermal Resistance	θ_{JA}	70	°C/W

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

READ/WRITE TIMING Figures 2 and 3, f = 1.0 MHz, Load Circuit of Figure 6.

Characteristic	Symbol	Min	Typ	Max	Unit
Address Delay	t_{AD}	–	220	300	ns
Peripheral Read Access Time $t_{acc} = t_{ut} - (t_{AD} + t_{DSR})$	t_{acc}	–	–	540	ns
Data Setup Time (Read)	t_{DSR}	100	–	–	ns
Input Data Hold Time	t_H	10	–	–	ns
Output Data Hold Time	t_H	10	25	–	ns
Address Hold Time (Address, R/W, VMA)	t_{AH}	50	75	–	ns
Enable High Time for DBE Input	t_{EH}	450	–	–	ns
Data Delay Time (Write)	t_{DDW}	–	165	225	ns
Processor Controls*					
Processor Control Setup Time	t_{PCS}	200	–	–	ns
Processor Control Rise and Fall Time	t_{PCr}, t_{PCf}	–	–	100	ns
Bus Available Delay	t_{BA}	–	–	300	ns
Three State Enable	t_{TSE}	–	–	40	ns
Three State Delay	t_{TSD}	–	–	700	ns
Data Bus Enable Down Time During $\phi 1$ Up Time (Figure 3)	$t_{\overline{DBE}}$	150	–	–	ns
Data Bus Enable Delay (Figure 3)	t_{DBED}	300	–	–	ns
Data Bus Enable Rise and Fall Times (Figure 3)	t_{DBEr}, t_{DBEf}	–	–	25	ns

*Additional information is given in Figures 12 through 16 of the Family Characteristics – see pages 17 through 20.

FIGURE 2 – READ DATA FROM MEMORY OR PERIPHERALS

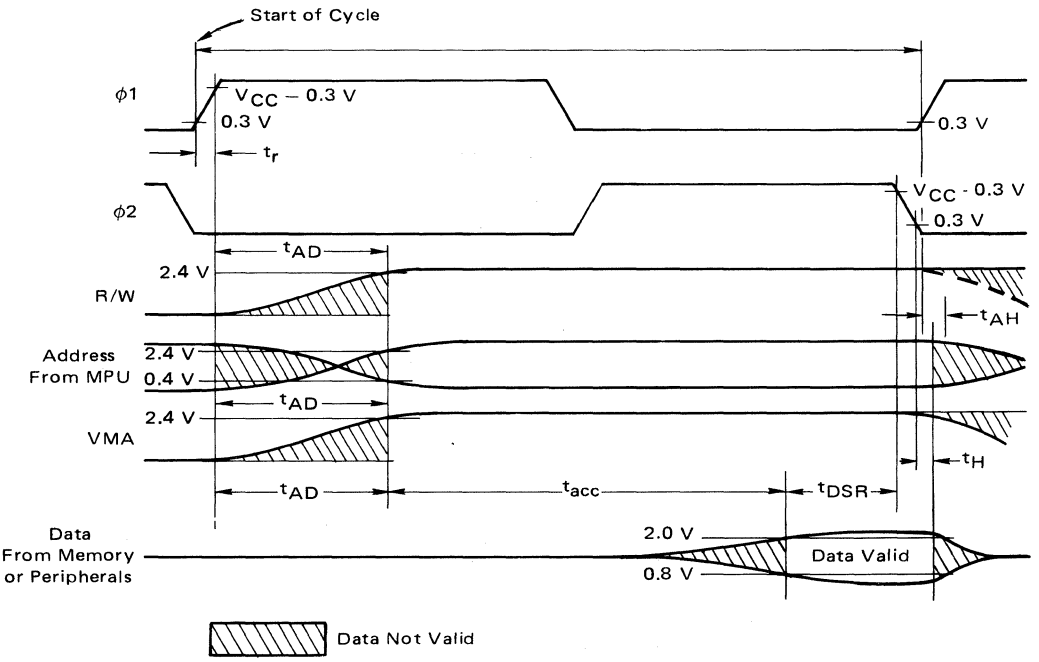


FIGURE 3 – WRITE IN MEMORY OR PERIPHERALS

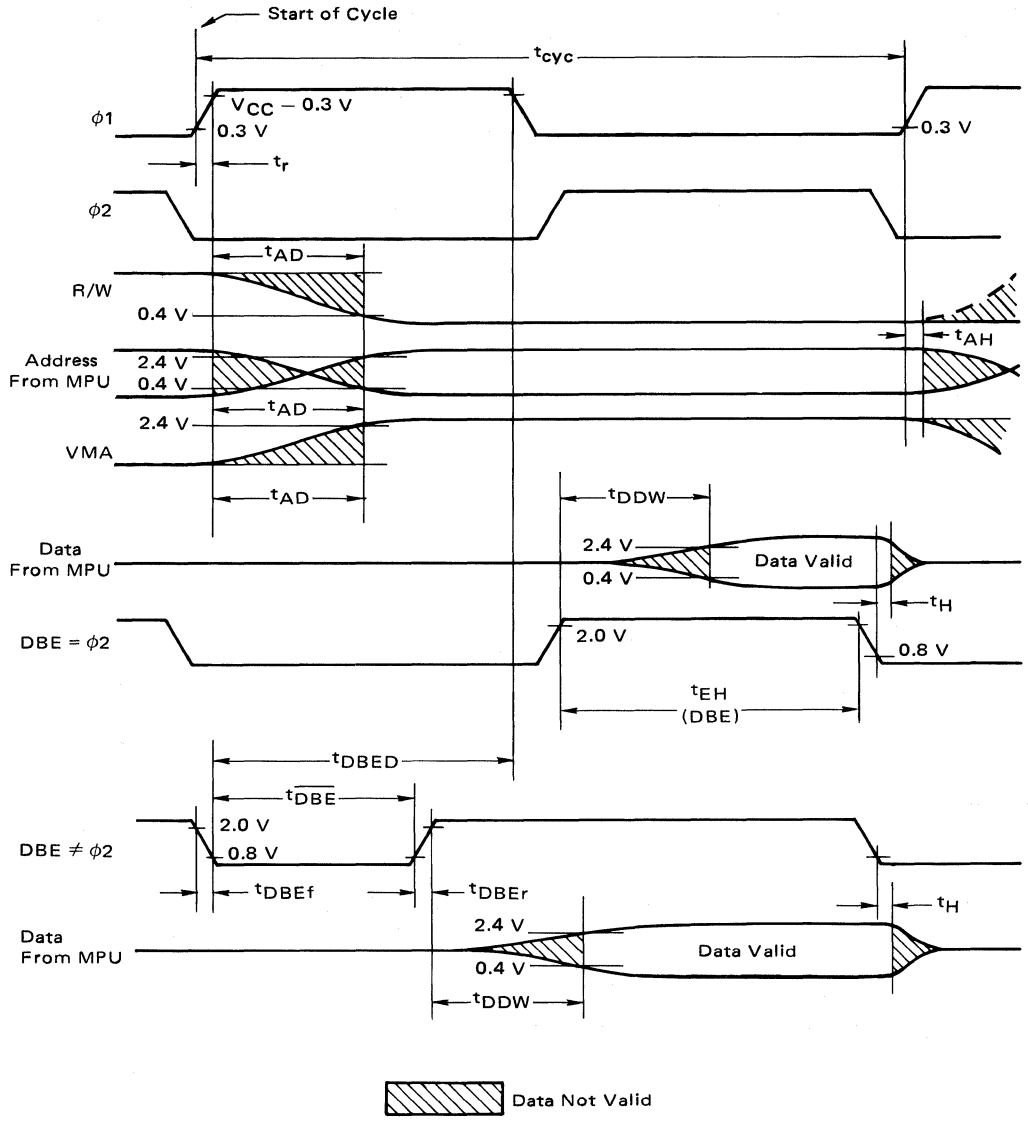


FIGURE 4 – TYPICAL DATA BUS OUTPUT DELAY versus CAPACITIVE LOADING

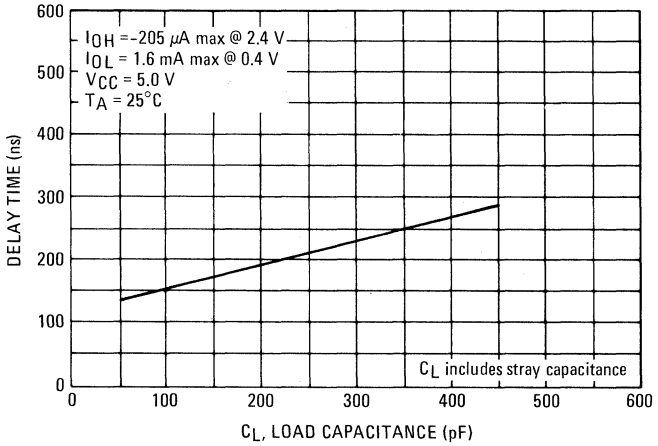


FIGURE 5 – TYPICAL READ/WRITE, VMA, AND ADDRESS OUTPUT DELAY versus CAPACITIVE LOADING

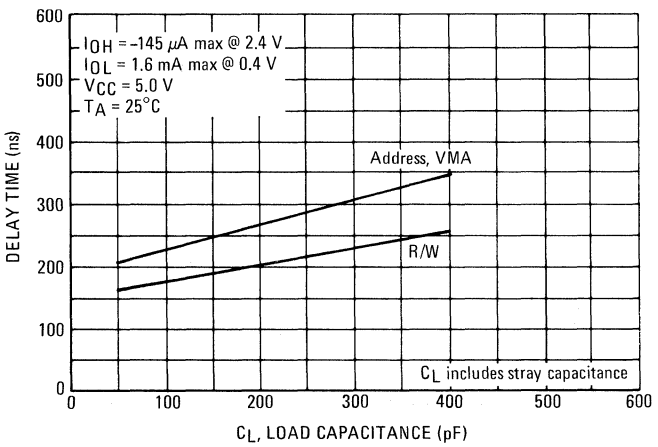
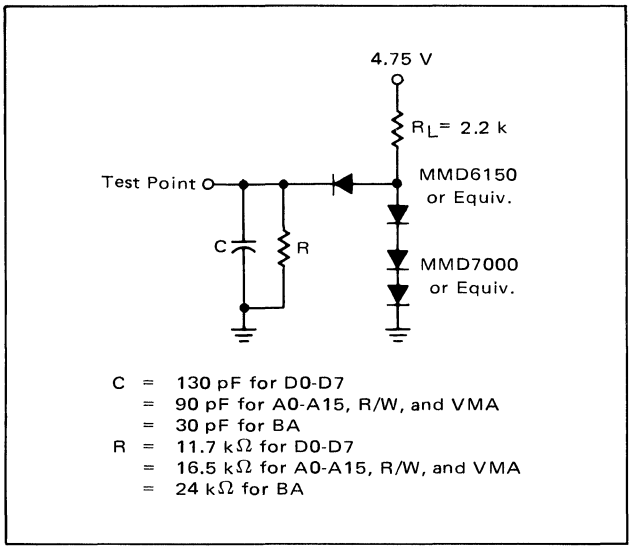


FIGURE 6 – BUS TIMING TEST LOAD



C = 130 pF for D0-D7
= 90 pF for A0-A15, R/W, and VMA
= 30 pF for BA
R = 11.7 kΩ for D0-D7
= 16.5 kΩ for A0-A15, R/W, and VMA
= 24 kΩ for BA

TYPICAL POWER SUPPLY CURRENT

FIGURE 7 – VARIATIONS WITH FREQUENCY

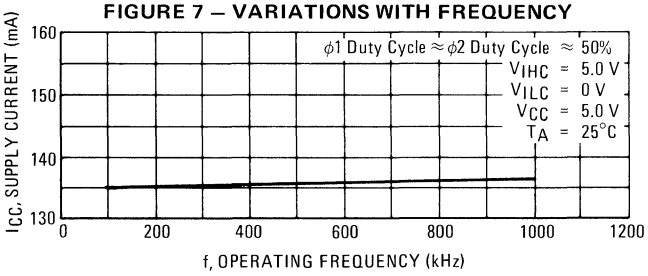
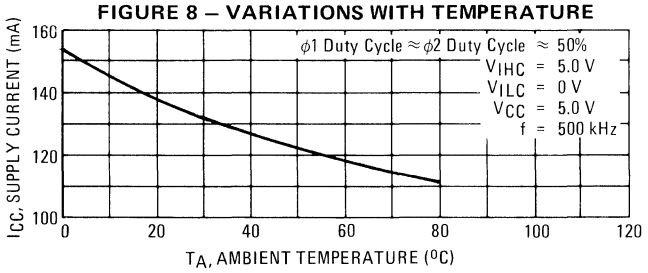
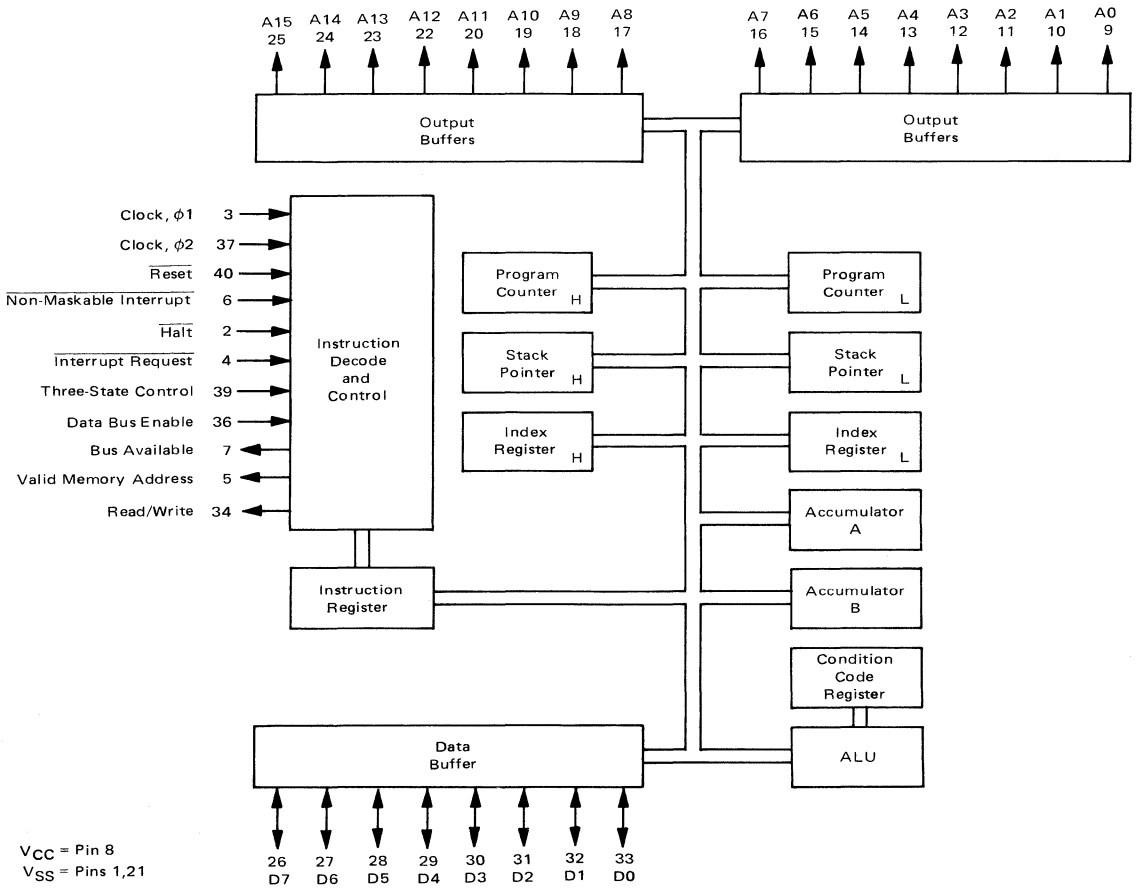


FIGURE 8 – VARIATIONS WITH TEMPERATURE



EXPANDED BLOCK DIAGRAM



MPU SIGNAL DESCRIPTION

Proper operation of the MPU requires that certain control and timing signals be provided to accomplish specific functions and that other signal lines be monitored to determine the state of the processor.

Clocks Phase One and Phase Two ($\phi 1, \phi 2$) — Two pins are used for a two-phase non-overlapping clock that runs at the V_{CC} voltage level.

Address Bus (A0-A15) — Sixteen pins are used for the address bus. The outputs are three-state bus drivers capable of driving one standard TTL load and 130 pF. When the output is turned off, it is essentially an open circuit. This permits the MPU to be used in DMA applications.

Data Bus (D0-D7) — Eight pins are used for the data bus. It is bi-directional, transferring data to and from the memory and peripheral devices. It also has three-state output buffers capable of driving one standard TTL load and 130 pF.

$\overline{\text{Halt}}$ — When this input is in the low state, all activity in the machine will be halted. This input is level sensitive. In the halt mode, the machine will stop at the end of an instruction, Bus Available will be at a one level, Valid Memory Address will be at a zero, and all other three-state lines will be in the three-state mode.

Transition of the $\overline{\text{Halt}}$ line must not occur during the last 250 ns of phase one. To insure single instruction operation, the $\overline{\text{Halt}}$ line must go high for one Clock cycle.

Three-State Control (TSC) — This input causes all of the address lines and the Read/Write line to go into the off or high impedance state. This state will occur 700 ns after TSC = 2.0 V. The Valid Memory Address and Bus Available signals will be forced low. The data bus is not affected by TSC and has its own enable (Data Bus Enable). In DMA applications, the Three-State Control line should be brought high on the leading edge of the Phase One Clock. The $\phi 1$ clock must be held in the high state and the $\phi 2$ in the low state for this function to operate properly. The address bus will then be available for other devices to directly address memory. Since the MPU is a dynamic device, it can be held in this state for only 4.5 μs or destruction of data will occur in the MPU.

Read/Write (R/W) — This TTL compatible output signals the peripherals and memory devices whether the MPU is in a Read (high) or Write (low) state. The normal standby state of this signal is Read (high). Three-State Control going high will turn Read/Write to the off (high impedance) state. Also, when the processor is halted, it will be in the off state. This output is capable of driving one standard TTL load and 90 pF.

Valid Memory Address (VMA) — This output indicates to peripheral devices that there is a valid address on the address bus. In normal operation, this signal should be utilized for enabling peripheral interfaces such as the PIA and ACIA. This signal is not three-state. One standard TTL load and 90 pF may be directly driven by this active high signal.

Data Bus Enable (DBE) — This input is the three-state control signal for the MPU data bus and will enable the bus drivers when in the high state. This input is TTL compatible; however in normal operation, it would be driven by the phase two clock. During an MPU read cycle, the data bus drivers will be disabled internally. When it is desired that another device control the data bus such as in Direct Memory Access (DMA) applications, DBE should be held low.

Bus Available (BA) — The Bus Available signal will normally be in the low state; when activated, it will go to the high state indicating that the microprocessor has stopped and that the address bus is available. This will occur if the $\overline{\text{Halt}}$ line is in the low state or the processor is in the WAIT state as a result of the execution of a WAIT instruction. At such time, all three-state output drivers will go to their off state and other outputs to their normally inactive level. The processor is removed from the WAIT state by the occurrence of a maskable (mask bit I = 0) or nonmaskable interrupt. This output is capable of driving one standard TTL load and 30 pF.

Interrupt Request ($\overline{\text{IRQ}}$) — This level sensitive input requests that an interrupt sequence be generated within the machine. The processor will wait until it completes the current instruction that is being executed before it recognizes the request. At that time, if the interrupt mask bit in the Condition Code Register is not set, the machine will begin an interrupt sequence. The Index Register, Program Counter, Accumulators, and Condition Code Register are stored away on the stack. Next the MPU will respond to the interrupt request by setting the interrupt mask bit high so that no further interrupts may occur. At the end of the cycle, a 16-bit address will be loaded that points to a vectoring address which is located in memory locations FFF8 and FFF9. An address loaded at these locations causes the MPU to branch to an interrupt routine in memory.

The $\overline{\text{Halt}}$ line must be in the high state for interrupts to be serviced. Interrupts will be latched internally while $\overline{\text{Halt}}$ is low.

The $\overline{\text{IRQ}}$ has a high impedance pullup device internal to the chip; however a 3 kΩ external resistor to V_{CC} should be used for wire-OR and optimum control of interrupts.

$\overline{\text{Reset}}$ — This input is used to reset and start the MPU from a power down condition, resulting from a power failure or an initial start-up of the processor. If a high level is detected on the input, this will signal the MPU to begin the restart sequence. This will start execution of a routine to initialize the processor from its reset condition. All the higher order address lines will be forced high. For the restart, the last two (FFFE, FFFF) locations in memory will be used to load the program that is addressed by the program counter. During the restart routine, the interrupt mask bit is set and must be reset before the MPU can be interrupted by $\overline{\text{IRQ}}$.

Figure 9 shows the initialization of the microprocessor after restart. $\overline{\text{Reset}}$ must be held low for at least eight clock periods after V_{CC} reaches 4.75 volts. If $\overline{\text{Reset}}$ goes high prior to the leading edge of $\phi 2$, on the next $\phi 1$ the first restart memory vector address (FFFE) will appear on the address lines. This location should contain the higher order eight bits to be stored into the program counter. Following, the next address FFFF should contain the lower order eight bits to be stored into the program counter.

Non-Maskable Interrupt ($\overline{\text{NMI}}$) – A low-going edge on this input requests that a non-mask-interrupt sequence be generated within the processor. As with the $\overline{\text{Interrupt Request}}$ signal, the processor will complete the current instruction that is being executed before it recognizes the $\overline{\text{NMI}}$ signal. The interrupt mask bit in the Condition Code Register has no effect on $\overline{\text{NMI}}$.

The Index Register, Program Counter, Accumulators, and Condition Code Register are stored away on the stack. At the end of the cycle, a 16-bit address will be loaded that points to a vectoring address which is located in memory locations FFFC and FFFD. An address loaded at these locations causes the MPU to branch to a non-maskable interrupt routine in memory.

$\overline{\text{NMI}}$ has a high impedance pullup resistor internal to the chip; however a 3 kΩ external resistor to V_{CC} should be used for wire-OR and optimum control of interrupts.

Inputs $\overline{\text{IRQ}}$ and $\overline{\text{NMI}}$ are hardware interrupt lines that are sampled during $\phi 2$ and will start the interrupt routine on the $\phi 1$ following the completion of an instruction.

Figure 10 is a flow chart describing the major decision paths and interrupt vectors of the microprocessor. Table 1 gives the memory map for interrupt vectors.

FIGURE 9 – INITIALIZATION OF MPU AFTER RESTART

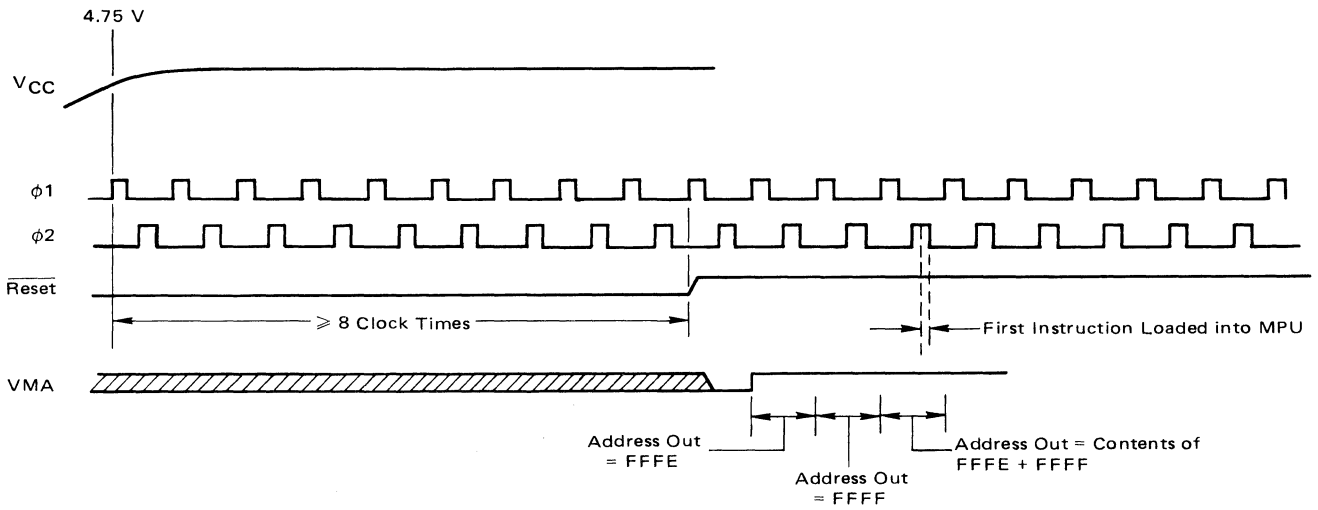
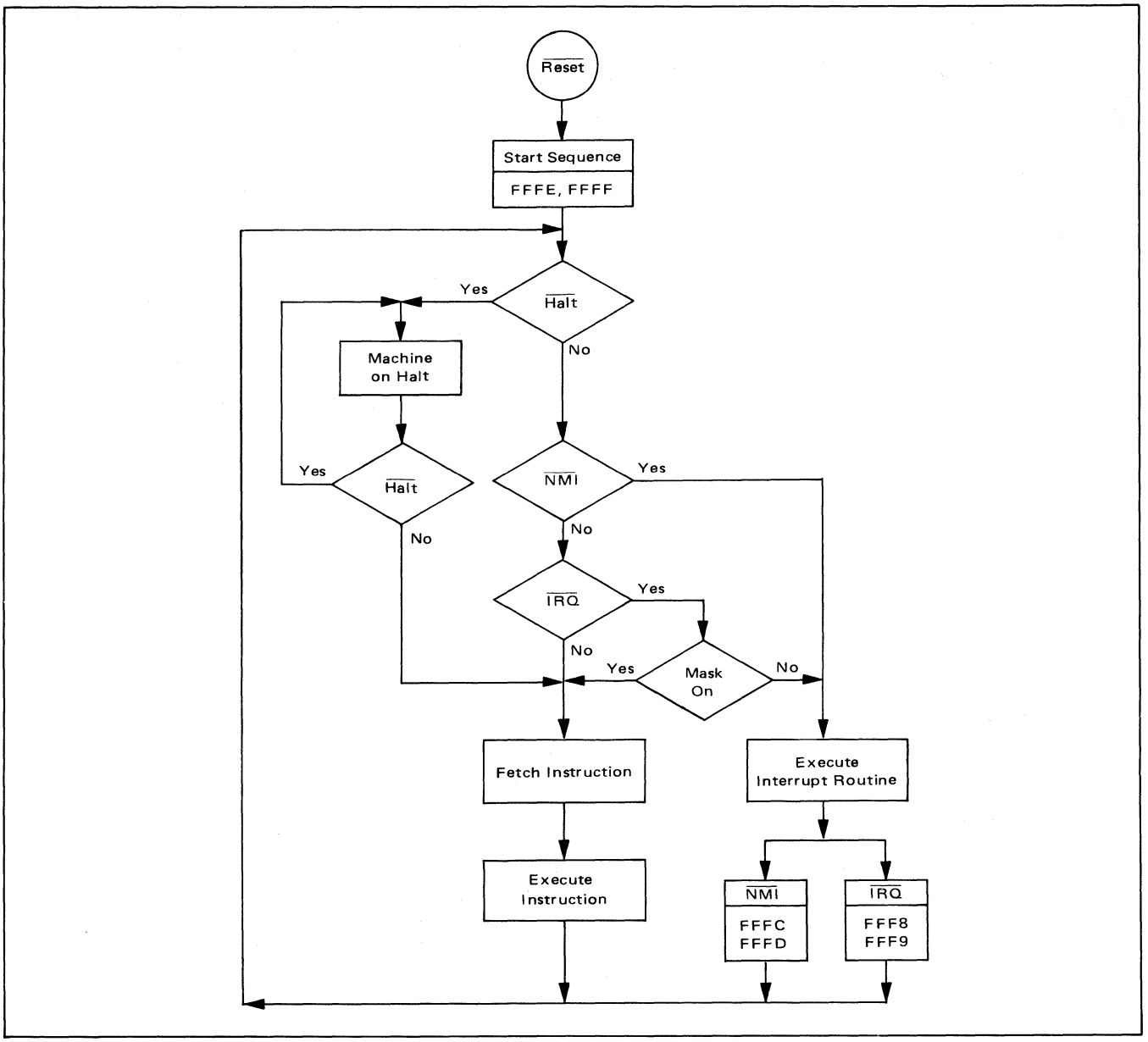


TABLE 1 – MEMORY MAP FOR INTERRUPT VECTORS

Vector MS	Vector LS	Description
FFFE	FFFF	Restart
FFFC	FFFD	Non-maskable Interrupt
FFFA	FFFB	Software Interrupt
FFF8	FFF9	Interrupt Request

FIGURE 10 – MPU FLOW CHART



MPU REGISTERS

The MPU has three 16-bit registers and three 8-bit registers available for use by the programmer (Figure 11).

Program Counter – The program counter is a two byte (16-bits) register that points to the current program address.

Stack Pointer – The stack pointer is a two byte register that contains the address of the next available location in an external push-down/pop-up stack. This stack is normally a random access Read/Write memory that may have any location (address) that is convenient. In those applications that require storage of information in the stack when power is lost, the stack must be non-volatile.

Index Register – The index register is a two byte register that is used to store data or a sixteen bit memory address for the Indexed mode of memory addressing.

Accumulators – The MPU contains two 8-bit accumulators that are used to hold operands and results from an arithmetic logic unit (ALU).

FIGURE 11 – PROGRAMMING MODEL OF THE MICROPROCESSING UNIT

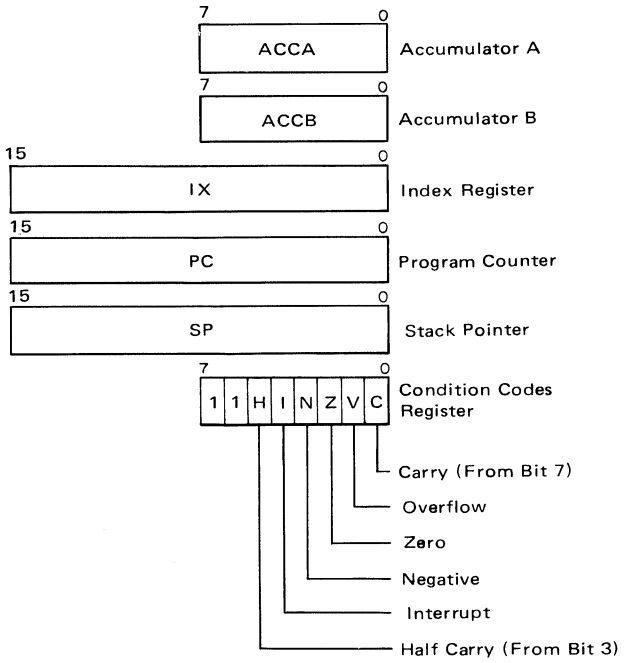
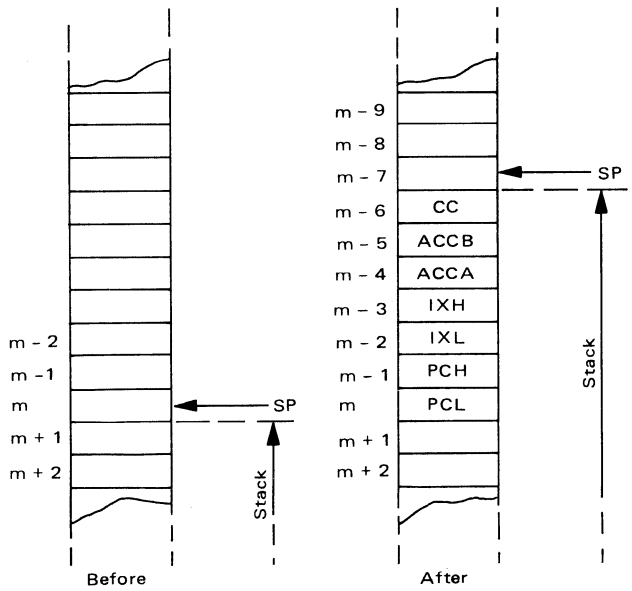


FIGURE 12 – SAVING THE STATUS OF THE MICROPROCESSOR IN THE STACK

SP = Stack Pointer
CC = Condition Codes (Also called the Processor Status Byte)
ACCB = Accumulator B
ACCA = Accumulator A
IXH = Index Register, Higher Order 8 Bits
IXL = Index Register, Lower Order 8 Bits
PCH = Program Counter, Higher Order 8 Bits
PCL = Program Counter, Lower Order 8 Bits



Condition Code Register — The condition code register indicates the results of an Arithmetic Logic Unit operation: Negative (N), Zero (Z), Overflow (V), Carry from bit 7 (C), and half carry from bit 3 (H). These bits of the Condition Code Register are used as testable conditions for the conditional branch instructions. Bit 4 is the interrupt mask bit (I). The unused bits of the Condition Code Register (b6 and b7) are ones.

Figure 12 shows the order of saving the microprocessor status within the stack.

MPU INSTRUCTION SET

The MC6800 has a set of 72 different instructions. Included are binary and decimal arithmetic, logical, shift, rotate, load, store, conditional or unconditional branch, interrupt and stack manipulation instructions (Tables 2 thru 6).

MPU ADDRESSING MODES

The MC6800 eight-bit microprocessing unit has seven address modes that can be used by a programmer, with the addressing mode a function of both the type of instruction and the coding within the instruction. A summary of the addressing modes for a particular instruction can be found in Table 7 along with the associated instruction execution time that is given in machine cycles. With a clock frequency of 1 MHz, these times would be microseconds.

Accumulator (ACCX) Addressing — In accumulator only addressing, either accumulator A or accumulator B is specified. These are one-byte instructions.

Immediate Addressing — In immediate addressing, the operand is contained in the second byte of the instruction except LDS and LDX which have the operand in the second and third bytes of the instruction. The MPU addresses this location when it fetches the immediate instruction for execution. These are two or three-byte instructions.

Direct Addressing — In direct addressing, the address of the operand is contained in the second byte of the instruction. Direct addressing allows the user to directly address the lowest 256 bytes in the machine i.e., locations zero through 255. Enhanced execution times are achieved by storing data in these locations. In most configurations, it should be a random access memory. These are two-byte instructions.

Extended Addressing — In extended addressing, the address contained in the second byte of the instruction is used as the higher eight-bits of the address of the operand. The third byte of the instruction is used as the lower eight-bits of the address for the operand. This is an absolute address in memory. These are three-byte instructions.

Indexed Addressing — In indexed addressing, the address contained in the second byte of the instruction is added to the index register's lowest eight bits in the MPU. The carry is then added to the higher order eight bits of the index register. This result is then used to address memory. The modified address is held in a temporary address register so there is no change to the index register. These are two-byte instructions.

Implied Addressing — In the implied addressing mode the instruction gives the address (i.e., stack pointer, index register, etc.). These are one-byte instructions.

Relative Addressing — In relative addressing, the address contained in the second byte of the instruction is added to the program counter's lowest eight bits plus two. The carry or borrow is then added to the high eight bits. This allows the user to address data within a range of -125 to +129 bytes of the present instruction. These are two-byte instructions.

TABLE 2 — MICROPROCESSOR INSTRUCTION SET — ALPHABETIC SEQUENCE

ABA	Add Accumulators
ADC	Add with Carry
ADD	Add
AND	Logical And
ASL	Arithmetic Shift Left
ASR	Arithmetic Shift Right
BCC	Branch if Carry Clear
BCS	Branch if Carry Set
BEQ	Branch if Equal to Zero
BGE	Branch if Greater or Equal Zero
BGT	Branch if Greater than Zero
BHI	Branch if Higher
BIT	Bit Test
BLE	Branch if Less or Equal
BLS	Branch if Lower or Same
BLT	Branch if Less than Zero
BMI	Branch if Minus
BNE	Branch if Not Equal to Zero
BPL	Branch if Plus
BRA	Branch Always
BSR	Branch to Subroutine
BVC	Branch if Overflow Clear
BVS	Branch if Overflow Set
CBA	Compare Accumulators
CLC	Clear Carry
CLI	Clear Interrupt Mask
CLR	Clear
CLV	Clear Overflow
CMP	Compare
COM	Complement
CPX	Compare Index Register
DAA	Decimal Adjust
DEC	Decrement
DES	Decrement Stack Pointer
DEX	Decrement Index Register
EOR	Exclusive OR
INC	Increment
INS	Increment Stack Pointer
INX	Increment Index Register
JMP	Jump
JSR	Jump to Subroutine
LDA	Load Accumulator
LDS	Load Stack Pointer
LDX	Load Index Register
LSR	Logical Shift Right
NEG	Negate
NOP	No Operation
ORA	Inclusive OR Accumulator
PSH	Push Data
PUL	Pull Data
ROL	Rotate Left
ROR	Rotate Right
RTI	Return from Interrupt
RTS	Return from Subroutine
SBA	Subtract Accumulators
SBC	Subtract with Carry
SEC	Set Carry
SEI	Set Interrupt Mask
SEV	Set Overflow
STA	Store Accumulator
STS	Store Stack Register
STX	Store Index Register
SUB	Subtract
SWI	Software Interrupt
TAB	Transfer Accumulators
TAP	Transfer Accumulators to Condition Code Reg.
TBA	Transfer Accumulators
TPA	Transfer Condition Code Reg. to Accumulator
TST	Test
TSX	Transfer Stack Pointer to Index Register
TXS	Transfer Index Register to Stack Pointer
WAI	Wait for Interrupt

TABLE 3 – ACCUMULATOR AND MEMORY INSTRUCTIONS

OPERATIONS	MNEMONIC	IMMED OP	IMMED ~	IMMED =	DIRECT OP	DIRECT ~	DIRECT =	INDEX OP	INDEX ~	INDEX =	EXTND OP	EXTND ~	EXTND =	IMPLIED OP	IMPLIED ~	IMPLIED =	BOOLEAN/ARITHMETIC OPERATION (All register labels refer to contents)	5 H	4 I	3 N	2 Z	1 V	0 C
Add	ADDA	8B	2	2	9B	3	2	AB	5	2	BB	4	3				A + M → A	↕	•	↕	↕	↕	↕
	ADDB	CB	2	2	DB	3	2	EB	5	2	FB	4	3				B + M → B	↕	•	↕	↕	↕	↕
Add Acmltrs	ABA													1B	2	1	A + B → A	↕	•	↕	↕	↕	↕
Add with Carry	ADCA	89	2	2	99	3	2	A9	5	2	B9	4	3				A + M + C → A	↕	•	↕	↕	↕	↕
	ADCB	C9	2	2	D9	3	2	E9	5	2	F9	4	3				B + M + C → B	↕	•	↕	↕	↕	↕
And	ANDA	84	2	2	94	3	2	A4	5	2	B4	4	3				A · M → A	•	•	↕	↕	R	•
	ANDB	C4	2	2	D4	3	2	E4	5	2	F4	4	3				B · M → B	•	•	↕	↕	R	•
Bit Test	BITA	85	2	2	95	3	2	A5	5	2	B5	4	3				A · M	•	•	↕	↕	R	•
	BITB	C5	2	2	D5	3	2	E5	5	2	F5	4	3				B · M	•	•	↕	↕	R	•
Clear	CLR							6F	7	2	7F	6	3				00 → M	•	•	R	S	R	R
	CLRA													4F	2	1	00 → A	•	•	R	S	R	R
	CLRB													5F	2	1	00 → B	•	•	R	S	R	R
Compare	CMPA	81	2	2	91	3	2	A1	5	2	B1	4	3				A − M	•	•	↕	↕	↕	↕
	CMPB	C1	2	2	D1	3	2	E1	5	2	F1	4	3				B − M	•	•	↕	↕	↕	↕
Compare Acmltrs	CBA													11	2	1	A − B	•	•	↕	↕	↕	↕
Complement, 1's	COM							63	7	2	73	6	3				$\overline{M} \rightarrow M$	•	•	↕	↕	R	S
	COMA													43	2	1	$\overline{A} \rightarrow A$	•	•	↕	↕	R	S
	COMB													53	2	1	$\overline{B} \rightarrow B$	•	•	↕	↕	R	S
Complement, 2's (Negate)	NEG							60	7	2	70	6	3				00 − M → M	•	•	↕	↕	①	②
	NEGA													40	2	1	00 − A → A	•	•	↕	↕	①	②
	NEGB													50	2	1	00 − B → B	•	•	↕	↕	①	②
Decimal Adjust, A	DAA													19	2	1	Converts Binary Add. of BCD Characters into BCD Format	•	•	↕	↕	↕	③
Decrement	DEC							6A	7	2	7A	6	3				M − 1 → M	•	•	↕	↕	4	•
	DECA													4A	2	1	A − 1 → A	•	•	↕	↕	4	•
	DECB													5A	2	1	B − 1 → B	•	•	↕	↕	4	•
Exclusive OR	EORA	88	2	2	98	3	2	A8	5	2	B8	4	3				A⊕M → A	•	•	↕	↕	R	•
	EORB	C8	2	2	D8	3	2	E8	5	2	F8	4	3				B⊕M → B	•	•	↕	↕	R	•
Increment	INC							6C	7	2	7C	6	3				M + 1 → M	•	•	↕	↕	⑤	•
	INCA													4C	2	1	A + 1 → A	•	•	↕	↕	⑤	•
	INCB													5C	2	1	B + 1 → B	•	•	↕	↕	⑤	•
Load Acmltr	LDAA	86	2	2	96	3	2	A6	5	2	B6	4	3				M → A	•	•	↕	↕	R	•
	LDAB	C6	2	2	D6	3	2	E6	5	2	F6	4	3				M → B	•	•	↕	↕	R	•
Or, Inclusive	ORAA	8A	2	2	9A	3	2	AA	5	2	BA	4	3				A + M → A	•	•	↕	↕	R	•
	ORAB	CA	2	2	DA	3	2	EA	5	2	FA	4	3				B + M → B	•	•	↕	↕	R	•
Push Data	PSHA													36	4	1	$A \rightarrow M_{SP}$, SP − 1 → SP	•	•	•	•	•	•
	PSHB													37	4	1	$B \rightarrow M_{SP}$, SP − 1 → SP	•	•	•	•	•	•
Pull Data	PULA													32	4	1	SP + 1 → SP, $M_{SP} \rightarrow A$	•	•	•	•	•	•
	PULB													33	4	1	SP + 1 → SP, $M_{SP} \rightarrow B$	•	•	•	•	•	•
Rotate Left	ROL							69	7	2	79	6	3				M	•	•	↕	↕	⑥	↕
	ROLA													49	2	1	A	•	•	↕	↕	⑥	↕
	ROLB													59	2	1	B — C b7 b0	•	•	↕	↕	⑥	↕
Rotate Right	ROR							66	7	2	76	6	3				M	•	•	↕	↕	⑥	↕
	RORA													46	2	1	A	•	•	↕	↕	⑥	↕
	RORB													56	2	1	B — C b7 b0	•	•	↕	↕	⑥	↕
Shift Left, Arithmetic	ASL							68	7	2	78	6	3				M	•	•	↕	↕	⑥	↕
	ASLA													48	2	1	A — 0	•	•	↕	↕	⑥	↕
	ASLB													58	2	1	B — C b7 b0	•	•	↕	↕	⑥	↕
Shift Right, Arithmetic	ASR							67	7	2	77	6	3				M	•	•	↕	↕	⑥	↕
	ASRA													47	2	1	A	•	•	↕	↕	⑥	↕
	ASRB													57	2	1	B — b7 b0 C	•	•	↕	↕	⑥	↕
Shift Right, Logic	LSR							64	7	2	74	6	3				M	•	•	R	↕	⑥	↕
	LSRA													44	2	1	A — 0	•	•	R	↕	⑥	↕
	LSRB													54	2	1	B — b7 b0 C	•	•	R	↕	⑥	↕
Store Acmltr.	STAA				97	4	2	A7	6	2	B7	5	3				A → M	•	•	↕	↕	R	•
	STAB				D7	4	2	E7	6	2	F7	5	3				B → M	•	•	↕	↕	R	•
Subtract	SUBA	80	2	2	90	3	2	A0	5	2	B0	4	3				A − M → A	•	•	↕	↕	↕	↕
	SUBB	C0	2	2	D0	3	2	E0	5	2	F0	4	3				B − M → B	•	•	↕	↕	↕	↕
Subtract Acmltrs.	SBA													10	2	1	A − B → A	•	•	↕	↕	↕	↕
Subtr. with Carry	SBCA	82	2	2	92	3	2	A2	5	2	B2	4	3				A − M − C → A	•	•	↕	↕	↕	↕
	SBCB	C2	2	2	D2	3	2	E2	5	2	F2	4	3				B − M − C → B	•	•	↕	↕	↕	↕
Transfer Acmltrs	TAB													16	2	1	A → B	•	•	↕	↕	R	•
	TBA													17	2	1	B → A	•	•	↕	↕	R	•
Test, Zero or Minus	TST							6D	7	2	7D	6	3				M − 00	•	•	↕	↕	R	R
	TSTA													4D	2	1	A − 00	•	•	↕	↕	R	R
	TSTB													5D	2	1	B − 00	•	•	↕	↕	R	R
																		H	I	N	Z	V	C

ADDRESSING MODES

LEGEND:

OP	Operation Code (Hexadecimal);	+	Boolean Inclusive OR;
~	Number of MPU Cycles;	⊕	Boolean Exclusive OR;
=	Number of Program Bytes;	$\overline{M}$	Complement of M;
+	Arithmetic Plus;	→	Transfer Into;
−	Arithmetic Minus;	0	Bit = Zero;
·	Boolean AND;	00	Byte = Zero;
M_{SP}	Contents of memory location pointed to be Stack Pointer;		

Note – Accumulator addressing mode instructions are included in the column for IMPLIED addressing

CONDITION CODE SYMBOLS:

H	Half-carry from bit 3;
I	Interrupt mask
N	Negative (sign bit)
Z	Zero (byte)
V	Overflow, 2's complement
C	Carry from bit 7
R	Reset Always
S	Set Always
↕	Test and set if true, cleared otherwise
•	Not Affected

TABLE 4 – INDEX REGISTER AND STACK MANIPULATION INSTRUCTIONS

		IMMED			DIRECT			INDEX			EXTND			IMPLIED				COND. CODE REG.					
																		5	4	3	2	1	0
POINTER OPERATIONS	MNEMONIC	OP	~	#	OP	~	#	OP	~	#	OP	~	#	OP	~	#	BOOLEAN/ARITHMETIC OPERATION	H	I	N	Z	V	C
Compare Index Reg	CPX	8C	3	3	9C	4	2	AC	6	2	BC	5	3				$X_H - M, X_L - (M+1)$	•	•	⑦	↕	⑧	•
Decrement Index Reg	DEX													09	4	1	$X - 1 \rightarrow X$	•	•	•	↕	•	•
Decrement Stack Pntr	DES													34	4	1	$SP - 1 \rightarrow SP$	•	•	•	•	•	•
Increment Index Reg	INX													08	4	1	$X + 1 \rightarrow X$	•	•	•	↕	•	•
Increment Stack Pntr	INS													31	4	1	$SP + 1 \rightarrow SP$	•	•	•	•	•	•
Load Index Reg	LDX	CE	3	3	DE	4	2	EE	6	2	FE	5	3				$M \rightarrow X_H, (M+1) \rightarrow X_L$	•	•	⑨	↕	R	•
Load Stack Pntr	LDS	8E	3	3	9E	4	2	AE	6	2	BE	5	3				$M \rightarrow SP_H, (M+1) \rightarrow SP_L$	•	•	⑨	↕	R	•
Store Index Reg	STX				DF	5	2	EF	7	2	FF	6	3				$X_H \rightarrow M, X_L \rightarrow (M+1)$	•	•	⑨	↕	R	•
Store Stack Pntr	STS				9F	5	2	AF	7	2	BF	6	3				$SP_H \rightarrow M, SP_L \rightarrow (M+1)$	•	•	⑨	↕	R	•
Indx Reg → Stack Pntr	TXS													35	4	1	$X - 1 \rightarrow SP$	•	•	•	•	•	•
Stack Pntr → Indx Reg	TSX													30	4	1	$SP + 1 \rightarrow X$	•	•	•	•	•	•

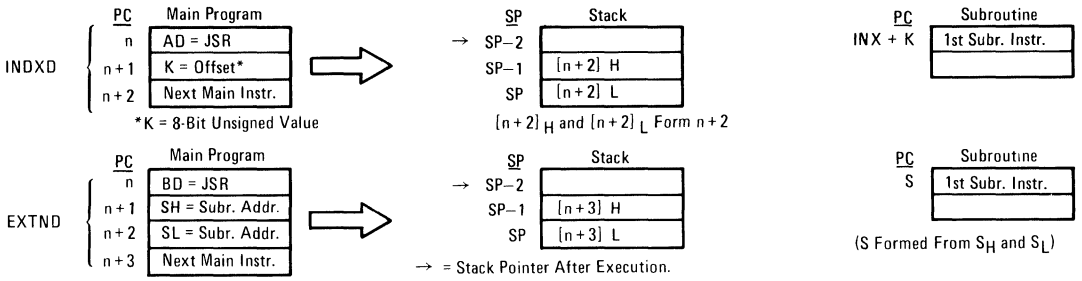
TABLE 5 – JUMP AND BRANCH INSTRUCTIONS

		RELATIVE			INDEX			EXTND			IMPLIED				COND. CODE REG.						
															5	4	3	2	1	0
OPERATIONS	MNEMONIC	OP	~	#	OP	~	#	OP	~	#	OP	~	#	BRANCH TEST	H	I	N	Z	V	C
Branch Always	BRA	20	4	2										None	•	•	•	•	•	•
Branch If Carry Clear	BCC	24	4	2										C = 0	•	•	•	•	•	•
Branch If Carry Set	BCS	25	4	2										C = 1	•	•	•	•	•	•
Branch If = Zero	BEQ	27	4	2										Z = 1	•	•	•	•	•	•
Branch If ≥ Zero	BGE	2C	4	2										$N \oplus V = 0$	•	•	•	•	•	•
Branch If > Zero	BGT	2E	4	2										$Z + (N \oplus V) = 0$	•	•	•	•	•	•
Branch If Higher	BHI	22	4	2										$C + Z = 0$	•	•	•	•	•	•
Branch If ≤ Zero	BLE	2F	4	2										$Z + (N \oplus V) = 1$	•	•	•	•	•	•
Branch If Lower Or Same	BLS	23	4	2										$C + Z = 1$	•	•	•	•	•	•
Branch If < Zero	BLT	2D	4	2										$N \oplus V = 1$	•	•	•	•	•	•
Branch If Minus	BMI	2B	4	2										N = 1	•	•	•	•	•	•
Branch If Not Equal Zero	BNE	26	4	2										Z = 0	•	•	•	•	•	•
Branch If Overflow Clear	BVC	28	4	2										V = 0	•	•	•	•	•	•
Branch If Overflow Set	BVS	29	4	2										V = 1	•	•	•	•	•	•
Branch If Plus	BPL	2A	4	2										N = 0	•	•	•	•	•	•
Branch To Subroutine	BSR	8D	8	2											•	•	•	•	•	•
Jump	JMP				6E	4	2	7E	3	3				See Special Operations	•	•	•	•	•	•
Jump To Subroutine	JSR				AD	8	2	BD	9	3					•	•	•	•	•	•
No Operation	NOP										01	2	1	Advances Prog. Cntr. Only	•	•	•	•	•	•
Return From Interrupt	RTI										3B	10	1		———			⑩		———
Return From Subroutine	RTS										39	5	1		•	•	•	•	•	•
Software Interrupt	SWI										3F	12	1	See Special Operations	•	•	•	•	•	•
Wait for Interrupt*	WAI										3E	9	1		•	⑪	•	•	•	•

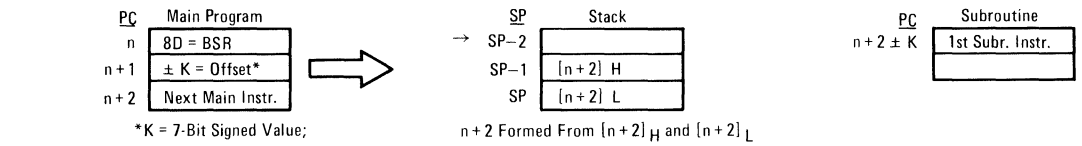
*WAI puts Address Bus, R/W, and Data Bus in the three-state mode while VMA is held low.

SPECIAL OPERATIONS

JSR, JUMP TO SUBROUTINE:

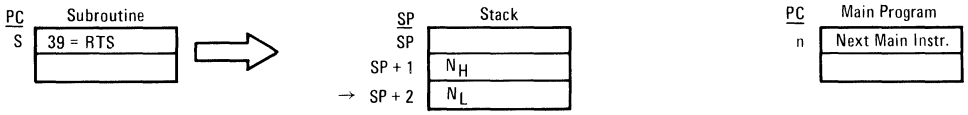


BSR, BRANCH TO SUBROUTINE:



JMP, JUMP:

RTS, RETURN FROM SUBROUTINE:



RTI, RETURN FROM INTERRUPT:

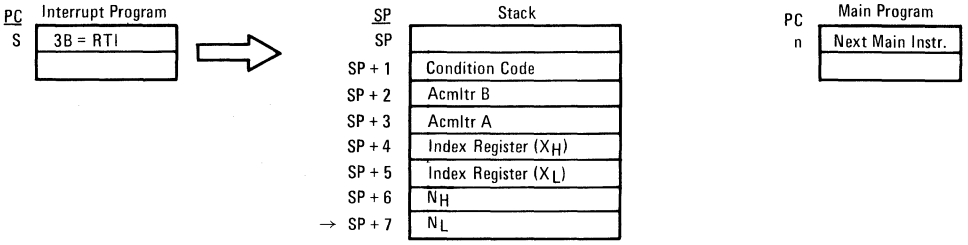


TABLE 6 – CONDITION CODE REGISTER MANIPULATION INSTRUCTIONS

OPERATIONS	MNEMONIC	IMPLIED OP	~	#	BOOLEAN OPERATION	COND. CODE REG. 5 H	4 I	3 N	2 Z	1 V	0 C
Clear Carry	CLC	0C	2	1	0 → C	•	•	•	•	•	R
Clear Interrupt Mask	CLI	0E	2	1	0 → I	•	R	•	•	•	•
Clear Overflow	CLV	0A	2	1	0 → V	•	•	•	•	R	•
Set Carry	SEC	0D	2	1	1 → C	•	•	•	•	•	S
Set Interrupt Mask	SEI	0F	2	1	1 → I	•	S	•	•	•	•
Set Overflow	SEV	0B	2	1	1 → V	•	•	•	•	S	•
Acmltr A → CCR	TAP	06	2	1	A → CCR	———			(12)		———
CCR → Acmltr A	TPA	07	2	1	CCR → A	•	•	•	•	•	•

CONDITION CODE REGISTER NOTES: (Bit set if test is true and cleared otherwise)

1 (Bit V) Test: Result = 10000000?
2 (Bit C) Test: Result = 00000000?
3 (Bit C) Test: Decimal value of most significant BCD Character greater than nine? (Not cleared if previously set.)
4 (Bit V) Test: Operand = 10000000 prior to execution?
5 (Bit V) Test: Operand = 01111111 prior to execution?
6 (Bit V) Test: Set equal to result of N⊕C after shift has occurred.
7 (Bit N) Test: Sign bit of most significant (MS) byte = 1?
8 (Bit V) Test: 2's complement overflow from subtraction of MS bytes?
9 (Bit N) Test: Result less than zero? (Bit 15 = 1)
10 (All) Load Condition Code Register from Stack. (See Special Operations)
11 (Bit I) Set when interrupt occurs. If previously set, a Non-Maskable Interrupt is required to exit the wait state.
12 (All) Set according to the contents of Accumulator A.

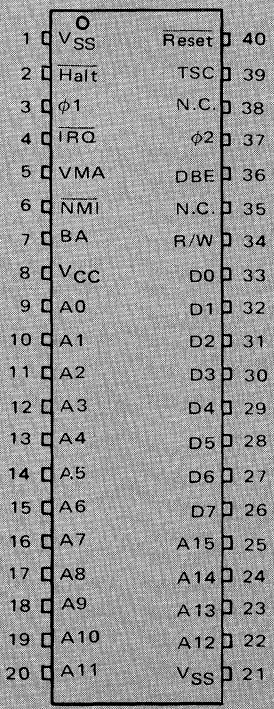
TABLE 7 – INSTRUCTION ADDRESSING MODES AND ASSOCIATED EXECUTION TIMES
(Times in Machine Cycles)

	(Dual Operand)	ACCX	Immediate	Direct	Extended	Indexed	Implied	Relative
ABA		•	•	•	•	•	2	•
ADC	x	•	2	3	4	5	•	•
ADD	x	•	2	3	4	5	•	•
AND	x	•	2	3	4	5	•	•
ASL		2	•	•	6	7	•	•
ASR		2	•	•	6	7	•	•
BCC		•	•	•	•	•	•	4
BCS		•	•	•	•	•	•	4
BEA		•	•	•	•	•	•	4
BGE		•	•	•	•	•	•	4
BGT		•	•	•	•	•	•	4
BHI		•	•	•	•	•	•	4
BIT	x	•	2	3	4	5	•	•
BLE		•	•	•	•	•	•	4
BLS		•	•	•	•	•	•	4
BLT		•	•	•	•	•	•	4
BMI		•	•	•	•	•	•	4
BNE		•	•	•	•	•	•	4
BPL		•	•	•	•	•	•	4
BRA		•	•	•	•	•	•	4
BSR		•	•	•	•	•	•	8
BVC		•	•	•	•	•	•	4
BVS		•	•	•	•	•	•	4
CBA		•	•	•	•	•	2	•
CLC		•	•	•	•	•	2	•
CLI		•	•	•	•	•	2	•
CLR		2	•	•	6	7	•	•
CLV		•	•	•	•	•	2	•
CMP	x	•	2	3	4	5	•	•
COM		2	•	•	6	7	•	•
CPX		•	3	4	5	6	•	•
DAA		•	•	•	•	•	2	•
DEC		2	•	•	6	7	•	•
DES		•	•	•	•	•	4	•
DEX		•	•	•	•	•	4	•
EOR	x	•	2	3	4	5	•	•

	(Dual Operand)	ACCX	Immediate	Direct	Extended	Indexed	Implied
INC		2	•	•	6	7	•
INS		•	•	•	•	•	4
INX		•	•	•	•	•	4
JMP		•	•	•	3	4	•
JSR		•	•	•	9	8	•
LDA	x	•	2	3	4	5	•
LDS		•	3	4	5	6	•
LDX		•	3	4	5	6	•
LSR		2	•	•	6	7	•
NEG		2	•	•	6	7	•
NOP		•	•	•	•	•	2
ORA	x	•	2	3	4	5	•
PSH		•	•	•	•	•	4
PUL		•	•	•	•	•	4
ROL		2	•	•	6	7	•
ROR		2	•	•	6	7	•
RTI		•	•	•	•	•	10
RTS		•	•	•	•	•	5
SBA		•	•	•	•	•	2
SBC	x	•	2	3	4	5	•
SEC		•	•	•	•	•	2
SEI		•	•	•	•	•	2
SEV		•	•	•	•	•	2
STA	x	•	•	4	5	6	•
STS		•	•	5	6	7	•
STX		•	•	5	6	7	•
SUB	x	•	2	3	4	5	•
SWI		•	•	•	•	•	12
TAB		•	•	•	•	•	2
TAP		•	•	•	•	•	2
TBA		•	•	•	•	•	2
TPA		•	•	•	•	•	2
TST		2	•	•	6	7	•
TSX		•	•	•	•	•	4
TSX		•	•	•	•	•	4
WAI		•	•	•	•	•	9

NOTE: Interrupt time is 12 cycles from the end of the instruction being executed, except following a WAI instruction. Then it is 4 cycles.

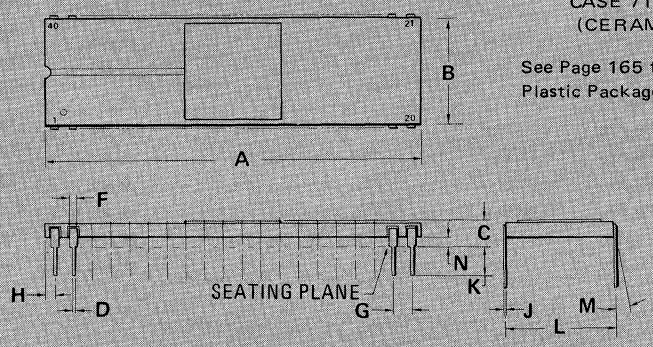
PIN ASSIGNMENT



PACKAGE DIMENSIONS

CASE 715-02
(CERAMIC)

See Page 165 for Plastic Package dimensions.



DIM	MILLIMETERS MIN	MILLIMETERS MAX	INCHES MIN	INCHES MAX
A	50.29	51.31	1.980	2.020
B	14.86	15.62	0.585	0.615
C	2.54	4.19	0.100	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.33	0.008	0.013
K	2.54	4.19	0.100	0.165
L	14.60	15.37	0.575	0.605
M	–	10^0	–	10^0
N	0.51	1.52	0.020	0.060

NOTE:
1. LEADS, TRUE POSITIONED WITHIN 0.25 mm (0.010) DIA (AT SEATING PLANE), AT MAX. MAT'L CONDITION.

SUMMARY OF CYCLE BY CYCLE OPERATION

Table 8 provides a detailed description of the information present on the Address Bus, Data Bus, Valid Memory Address line (VMA), and the Read/Write line (R/W) during each cycle for each instruction.

This information is useful in comparing actual with expected results during debug of both software and hardware as the control program is executed. The information is categorized in groups according to Addressing Mode and Number of Cycles per instruction. (In general, instructions with the same Addressing Mode and Number of Cycles execute in the same manner; exceptions are indicated in the table.)

TABLE 8 – OPERATION SUMMARY

Address Mode and Instructions	Cycles	Cycle #	VMA Line	Address Bus	R/W Line	Data Bus
IMMEDIATE						
ADC EOR ADD LDA AND ORA BIT SBC CMP SUB	2	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Operand Data
CPX LDS LDX	3	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Operand Data (High Order Byte)
		3	1	Op Code Address + 2	1	Operand Data (Low Order Byte)
DIRECT						
ADC EOR ADD LDA AND ORA BIT SBC CMP SUB	3	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand
		3	1	Address of Operand	1	Operand Data
CPX LDS LDX	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand
		3	1	Address of Operand	1	Operand Data (High Order Byte)
		4	1	Operand Address + 1	1	Operand Data (Low Order Byte)
STA	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Destination Address
		3	0	Destination Address	1	Irrelevant Data (Note 1)
		4	1	Destination Address	0	Data from Accumulator
STS STX	5	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand
		3	0	Address of Operand	1	Irrelevant Data (Note 1)
		4	1	Address of Operand	0	Register Data (High Order Byte)
		5	1	Address of Operand + 1	0	Register Data (Low Order Byte)
INDEXED						
JMP	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
ADC EOR ADD LDA AND ORA BIT SBC CMP SUB	5	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
		5	1	Index Register Plus Offset	1	Operand Data
CPX LDS LDX	6	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
		5	1	Index Register Plus Offset	1	Operand Data (High Order Byte)
		6	1	Index Register Plus Offset + 1	1	Operand Data (Low Order Byte)

TABLE 8 – OPERATION SUMMARY (Continued)

Address Mode and Instructions	Cycles	Cycle #	VMA Line	Address Bus	R/W Line	Data Bus
INDEXED (Continued)						
STA	6	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
		5	0	Index Register Plus Offset	1	Irrelevant Data (Note 1)
		6	1	Index Register Plus Offset	0	Operand Data
ASL LSR ASR NEG CLR ROL COM ROR DEC TST INC	7	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
		5	1	Index Register Plus Offset	1	Current Operand Data
		6	0	Index Register Plus Offset	1	Irrelevant Data (Note 1)
		7	1/0 (Note 3)	Index Register Plus Offset	0	New Operand Data (Note 3)
STS STX	7	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
		5	0	Index Register Plus Offset	1	Irrelevant Data (Note 1)
		6	1	Index Register Plus Offset	0	Operand Data (High Order Byte)
		7	1	Index Register Plus Offset + 1	0	Operand Data (Low Order Byte)
JSR	8	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Offset
		3	0	Index Register	1	Irrelevant Data (Note 1)
		4	1	Stack Pointer	0	Return Address (Low Order Byte)
		5	1	Stack Pointer – 1	0	Return Address (High Order Byte)
		6	0	Stack Pointer – 2	1	Irrelevant Data (Note 1)
		7	0	Index Register	1	Irrelevant Data (Note 1)
		8	0	Index Register Plus Offset (w/o Carry)	1	Irrelevant Data (Note 1)
EXTENDED						
JMP	3	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Jump Address (High Order Byte)
		3	1	Op Code Address + 2	1	Jump Address (Low Order Byte)
ADC EOR ADD LDA AND ORA BIT SBC CMP SUB	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand (High Order Byte)
		3	1	Op Code Address + 2	1	Address of Operand (Low Order Byte)
		4	1	Address of Operand	1	Operand Data
CPX LDS LDX	5	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand (High Order Byte)
		3	1	Op Code Address + 2	1	Address of Operand (Low Order Byte)
		4	1	Address of Operand	1	Operand Data (High Order Byte)
		5	1	Address of Operand + 1	1	Operand Data (Low Order Byte)
STA A STA B	5	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Destination Address (High Order Byte)
		3	1	Op Code Address + 2	1	Destination Address (Low Order Byte)
		4	0	Operand Destination Address	1	Irrelevant Data (Note 1)
		5	1	Operand Destination Address	0	Data from Accumulator
ASL LSR ASR NEG CLR ROL COM ROR DEC TST INC	6	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand (High Order Byte)
		3	1	Op Code Address + 2	1	Address of Operand (Low Order Byte)
		4	1	Address of Operand	1	Current Operand Data
		5	0	Address of Operand	1	Irrelevant Data (Note 1)
		6	1/0 (Note 3)	Address of Operand	0	New Operand Data (Note 3)

TABLE 8 – OPERATION SUMMARY (Continued)

Address Mode and Instructions	Cycles	Cycle #	VMA Line	Address Bus	R/W Line	Data Bus
EXTENDED (Continued)						
STS STX	6	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Operand (High Order Byte)
		3	1	Op Code Address + 2	1	Address of Operand (Low Order Byte)
		4	0	Address of Operand	1	Irrelevant Data (Note 1)
		5	1	Address of Operand	0	Operand Data (High Order Byte)
		6	1	Address of Operand + 1	0	Operand Data (Low Order Byte)
JSR	9	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Address of Subroutine (High Order Byte)
		3	1	Op Code Address + 2	1	Address of Subroutine (Low Order Byte)
		4	1	Subroutine Starting Address	1	Op Code of Next Instruction
		5	1	Stack Pointer	0	Return Address (Low Order Byte)
		6	1	Stack Pointer – 1	0	Return Address (High Order Byte)
		7	0	Stack Pointer – 2	1	Irrelevant Data (Note 1)
		8	0	Op Code Address + 2	1	Irrelevant Data (Note 1)
		9	1	Op Code Address + 2	1	Address of Subroutine (Low Order Byte)
INHERENT						
ABA DAA SEC ASL DEC SEI ASR INC SEV CBA LSR TAB CLC NEG TAP CLI NOP TBA CLR ROL TPA CLV ROR TST COM SBA	2	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
DES DEX INS INX	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	0	Previous Register Contents	1	Irrelevant Data (Note 1)
		4	0	New Register Contents	1	Irrelevant Data (Note 1)
PSH	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	1	Stack Pointer	0	Accumulator Data
		4	0	Stack Pointer – 1	1	Accumulator Data
PUL	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	0	Stack Pointer	1	Irrelevant Data (Note 1)
		4	1	Stack Pointer + 1	1	Operand Data from Stack
TSX	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	0	Stack Pointer	1	Irrelevant Data (Note 1)
		4	0	New Index Register	1	Irrelevant Data (Note 1)
TXS	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	0	Index Register	1	Irrelevant Data
		4	0	New Stack Pointer	1	Irrelevant Data
RTS	5	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Irrelevant Data (Note 2)
		3	0	Stack Pointer	1	Irrelevant Data (Note 1)
		4	1	Stack Pointer + 1	1	Address of Next Instruction (High Order Byte)
		5	1	Stack Pointer + 2	1	Address of Next Instruction (Low Order Byte)

TABLE 8 – OPERATION SUMMARY (Continued)

Address Mode and Instructions	Cycles	Cycle #	VMA Line	Address Bus	R/W Line	Data Bus
INHERENT (Continued)						
WAI	9	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Op Code of Next Instruction
		3	1	Stack Pointer	0	Return Address (Low Order Byte)
		4	1	Stack Pointer – 1	0	Return Address (High Order Byte)
		5	1	Stack Pointer – 2	0	Index Register (Low Order Byte)
		6	1	Stack Pointer – 3	0	Index Register (High Order Byte)
		7	1	Stack Pointer – 4	0	Contents of Accumulator A
		8	1	Stack Pointer – 5	0	Contents of Accumulator B
		9	1	Stack Pointer – 6 (Note 4)	1	Contents of Cond. Code Register
RTI	10	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Irrelevant Data (Note 2)
		3	0	Stack Pointer	1	Irrelevant Data (Note 1)
		4	1	Stack Pointer + 1	1	Contents of Cond. Code Register from Stack
		5	1	Stack Pointer + 2	1	Contents of Accumulator B from Stack
		6	1	Stack Pointer + 3	1	Contents of Accumulator A from Stack
		7	1	Stack Pointer + 4	1	Index Register from Stack (High Order Byte)
		8	1	Stack Pointer + 5	1	Index Register from Stack (Low Order Byte)
		9	1	Stack Pointer + 6	1	Next Instruction Address from Stack (High Order Byte)
		10	1	Stack Pointer + 7	1	Next Instruction Address from Stack (Low Order Byte)
SWI	12	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Irrelevant Data (Note 1)
		3	1	Stack Pointer	0	Return Address (Low Order Byte)
		4	1	Stack Pointer – 1	0	Return Address (High Order Byte)
		5	1	Stack Pointer – 2	0	Index Register (Low Order Byte)
		6	1	Stack Pointer – 3	0	Index Register (High Order Byte)
		7	1	Stack Pointer – 4	0	Contents of Accumulator A
		8	1	Stack Pointer – 5	0	Contents of Accumulator B
		9	1	Stack Pointer – 6	0	Contents of Cond. Code Register
		10	0	Stack Pointer – 7	1	Irrelevant Data (Note 1)
		11	1	Vector Address FFFA (Hex)	1	Address of Subroutine (High Order Byte)
		12	1	Vector Address FFFB (Hex)	1	Address of Subroutine (Low Order Byte)
RELATIVE						
BCC BHI BNE BCS BLE BPL BEQ BLS BRA BGE BLT BVC BGT BMI BVS	4	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Branch Offset
		3	0	Op Code Address + 2	1	Irrelevant Data (Note 1)
		4	0	Branch Address	1	Irrelevant Data (Note 1)
BSR	8	1	1	Op Code Address	1	Op Code
		2	1	Op Code Address + 1	1	Branch Offset
		3	0	Return Address of Main Program	1	Irrelevant Data (Note 1)
		4	1	Stack Pointer	0	Return Address (Low Order Byte)
		5	1	Stack Pointer – 1	0	Return Address (High Order Byte)
		6	0	Stack Pointer – 2	1	Irrelevant Data (Note 1)
		7	0	Return Address of Main Program	1	Irrelevant Data (Note 1)
		8	0	Subroutine Address	1	Irrelevant Data (Note 1)

Note 1. If device which is addressed during this cycle uses VMA, then the Data Bus will go to the high impedance three-state condition. Depending on bus capacitance, data from the previous cycle may be retained on the Data Bus.
Note 2. Data is ignored by the MPU.
Note 3. For TST, VMA = 0 and Operand data does not change.
Note 4. While the MPU is waiting for the interrupt, Bus Available will go high indicating the following states of the control lines: VMA is low; Address Bus, R/W, and Data Bus are all in the high impedance state.